RESTAURANT ANALYSIS

Identifying and Recommending Best Restaurants (Marketing Domain)

1. Problem Statement

A restaurant consolidator aims to revamp its B2C platform using intelligent automation. The objective is to identify key metrics and understand patterns in restaurant data to build an effective recommendation system. This requires analyzing customer preferences, restaurant characteristics, and factors influencing ratings.

2. Dataset Overview

- Total Records: **9,551 restaurants**
- Total Columns: **20 attributes**
- Data includes: Ratings, cost, cuisines, votes, delivery options, table booking, city, country

3. Data Preparation & Cleaning

Data preprocessing was performed using Excel and Power Query:

- No duplicate records found
- Missing values: Only 8 missing entries in cuisine column (negligible)
- Standardized inconsistent city names (e.g., Istanbul, Brasilia)
- Removed extra spaces and cleaned text fields
- Converted cost into INR using exchange rates
- Normalized cuisine data using unpivoting

4. Exploratory Data Analysis (EDA)

❖ **Overall Metrics**

- Total Restaurants: **9,551**
- Average Rating: **2.67**
- Average Cost for Two: **₹1199**

⇨ The overall rating indicates **moderate customer satisfaction**

❖ Geographic Distribution

- **India** has the highest number of restaurants (**8,652**)
- **New Delhi** has the maximum restaurant concentration
- Several cities have only **1 restaurant**, indicating uneven market distribution

❖ Ratings Distribution

- Maximum rating observed: **4.8 (25 restaurants)**
- Most common rating: **3.2 (522 restaurants)**
- Minimum rating: **1.8 (1 restaurant)**

⇨ Ratings are concentrated around the mid-range, showing average performance across restaurants

❖ Franchise Analysis
Top franchises with highest national presence:

- Café Coffee Day
- Domino's Pizza
- Subway
- McDonald's
- Green Chick Chop

⇨ These brands demonstrate strong scalability and wide geographic reach

❖ Table Booking vs No Table Booking

- **88% restaurants do not provide table booking**

⇨ Table booking is not widely adopted across the dataset

❖ Online Delivery Analysis

- Only **25% restaurants offer online delivery**

⇨ Delivery services are limited but critical

❖ Votes vs Delivery

- Restaurants offering online delivery receive **higher total and average votes**

⇨ Delivery-enabled restaurants have better customer engagement

❖ **Cuisine Analysis**

- Top cuisines identified across cities
- Maximum cuisines per restaurant: **8**
- Minimum cuisines per restaurant: **1**

Cuisine Count vs Rating Insight

- **1–2 cuisines** → lower ratings
- **4–6 cuisines** → highest ratings
- **7–8 cuisines** → slight decline in ratings

⇨ Moderate variety provides the best customer experience

❖ **Cost vs Rating**

- Lower-cost restaurants tend to have **higher ratings**

⇨ High price does not guarantee customer satisfaction

5. Factors Affecting Restaurant Ratings

The following variables significantly influence ratings:

1. Number of Cuisines

- Optimal range: **4–6 cuisines**
- Too few or too many negatively impact ratings

2. Cost

- Affordable restaurants receive better ratings
- Indicates importance of **value for money**

3. Online Delivery

- Strong positive impact on **customer engagement (votes)**

- Improves accessibility and convenience

4. Customer Votes

- Higher votes indicate popularity and trust
- Positively associated with better ratings

5. Table Booking

- Minimal impact compared to delivery

These variables—cost, cuisines, delivery, and votes—can be used to predict restaurant ratings and build a recommendation system.

6. Ideal Restaurant Profile (Recommendation Model)

Based on the analysis, high-performing restaurants typically have:

- Rating above 4.0
- Moderate pricing
- 3–6 cuisines
- Online delivery enabled
- High customer engagement (votes)

7. Final Recommendations

To improve the platform:

- Promote restaurants with high ratings and strong engagement
- Encourage adoption of online delivery services
- Recommend restaurants with balanced cuisine variety
- Highlight affordable, high-quality options
- Expand into cities with low restaurant presence

8. Conclusion

The analysis shows that restaurant success is primarily driven by **affordability, optimal menu variety, and service accessibility**, rather than premium pricing or excessive options. These insights can be effectively used to build a data-driven recommendation system and improve customer experience.

9. Tools & Technologies Used

- Excel (Data Cleaning)
- Power Query (Transformation)
- SQL (Data Analysis)
- Power BI (Visualization & Dashboarding)

10. Appendix (SQL Queries & Outputs)

❖ Total Restaurants

```
select count(distinct restaurant_id) as Total_Restaurant from restaurant_db;
```

Total_Restaurant
9551

❖ Average Ratings

```
select round(avg(rating), 2) as Avg_Rating from restaurant_db;
```

Avg_Rating
2.67

❖ City Distribution

```
select city, count(*) as Total_Restaurant
from restaurant_db
group by city
order by Total_Restaurant desc ;
```

Result Grid | Filter Rows:

city	Total_Restaurant
New Delhi	5473
Gurgaon	1118
Noida	1080
Faridabad	251
Ghaziabad	25
Lucknow	21
Ahmedabad	21
Guwahati	Ahmedabad
Bhubaneshwar	21
Amritsar	21
Agra	20
Kanpur	20
Jaipur	20

city	Total_Restaurant
Panchkula	1
Mohali	1
Paynesville	1
Flaxton	1
Middleton Beach	1
Montville	1
Palm Cove	1
Tanunda	1
Armidale	1
Balingup	1
Beechworth	1
East Ballina	1
Forrest	1

❖ Rating Distribution

```
select rating , count(restaurant_id) as Total_restaurant
from restaurant_db
group by rating
order by rating desc;
```

rating	Total_restaurant
4.8	25
4.7	42
4.6	78
4.5	95
4.4	144
4.3	174
4.2	221
4.1	274
4	266
3.9	335
3.8	400
3.7	427
3.6	458

rating	Total_restaurant
3.2	522
3.1	519
3.4	498
3.3	483
3.5	480
3	468
3.6	458
3.7	427
3.8	400
2.9	381
3.9	335
2.8	315
4.1	274

❖ Franchise Analysis

```
select restaurant_name, count(country_name) as total_country
from restaurant_db
group by restaurant_name
order by total_country desc ;
```

restaurant_name	total_country
Hard Rock Cafe	3
Applebee's	2
Barbeque Nation	2
Chili's Grill & Bar	2
Din Tai Fung	2
Farzi Cafe	2
Gazebo	2
Gymkhana	2
Ichiban	2
Indian Coffee House	2
Jamie's Italian	2
Nando's	2
Pizza Hut	2
Punjab Grill	2

❖ Delivery vs Votes

```
select has_online_delivery, count(*) as total_restaurant, sum(votes) as total_votes,
avg(votes) as avg_votes
from restaurant_db
```

has_online_delivery	total_restaurant	total_votes	avg_votes
No	7100	980731	138.1311
Yes	2451	517914	211.3072

❖ Cost vs Rating

```
select avg_cost_in_inr, round(Avg(rating)) as avg_rating
from restaurant_db
group by avg_cost_in_inr
order by avg_cost_in_inr;
```

avg_cost_in_inr	avg_rating
78	5
104	5
338	5
825	5
1035	5
1445	5
2453	5
3616	5
4068	5
4294	5
4520	5
5750	5
5876	5
6102	5